CHAN YI HENG

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Education

National University of Singapore — Singapore, SG
Master of Science in Statistics — *Aug 2026 – Expected 2027*

Singapore University of Social Sciences — Singapore, SG
Bachelor of Science (Honours) in Mathematics — *Jan 2022 – Dec 2025*

Singapore Polytechnic — Singapore, SG
Diploma in Banking & Finance — *Apr 2016 – Mar 2019*

Experience

Research Assistant — Oct 2025 – Present
Singapore University of Social Sciences, University Research Office — *Singapore, SG*

- Conducted methodological research at the intersection of statistical inference, causal learning, and computational statistics, with applications in the social and behavioral sciences.
- Developed statistical modeling frameworks incorporating heterogeneous treatment effects, robust estimation, and complex dependence structures.
- Investigated statistical learning and stochastic modeling methodologies, emphasizing non-stationarity, model specification, and inferential robustness.
- Designed reproducible computational workflows for statistical estimation, numerical experimentation, and empirical validation.

Teaching Assistant — Jan 2026 – Present
Singapore University of Social Sciences — *Singapore, SG*

- Facilitated undergraduate instruction in probability theory, statistical inference, asymptotic theory, and classical linear regression.
- Guided mathematical derivations, statistical problem-solving, and regression diagnostics, emphasizing theoretical foundations and model assumptions.

Research Intern — Nov 2024 – Aug 2025
*A*STAR, Advanced Remanufacturing and Technology Centre* — *Singapore, SG*

- Developed an end-to-end non-parametric learning framework for inventory optimization under stochastic demand, uncertain lead times, and non-stationary operating conditions.
- Integrated predictive representation learning with decision optimization, investigating alternatives to conventional predict-then-optimize formulations.
- Implemented explainable AI methodologies and conducted systematic benchmarking to evaluate decision quality, model robustness, and sensitivity to distributional shift.

Volunteer R&D Engineer — Nov 2023 – Present
Zaunkoenig GmbH — *Remote, International*

- Collaborated with an international engineering team across Germany, the UK, the USA, and Canada on technical prototyping, product development, and design optimization.
- Contributed to cable engineering and mechanical component optimization, balancing mass reduction, structural integrity, and functional performance.
- Coordinated regional technical operations across Southeast Asia, integrating engineering assessments and user feedback into iterative product refinements.

Finance Operations Analyst — Sep 2021 – Dec 2021
Indeed — *Singapore, SG*

- Performed financial data validation, reconciliation, and quality assurance to maintain data integrity across international operations.

Finance Specialist — Aug 2019 – Aug 2021
Singapore Armed Forces — *Singapore, SG*

- Oversaw financial operations, budgetary controls, and compliance assessments within government regulatory frameworks.

Wealth Management Intern — Sep 2018 – May 2019
DBS Bank — *Singapore, SG*

- Conducted operational analyses and systematic documentation reviews to support process efficiency and regulatory compliance.

Research

Regime-Aware Statistical Learning for Financial Markets | *Dr. Fan Zengyan*

- Developing a regime-aware statistical learning framework to characterize heterogeneous asset-price responses to exogenous macroeconomic shocks under structural breaks and time-varying dependence.
- Integrating high-frequency financial observations with macroeconomic state variables to model non-linear interactions, evolving dependence structures, and regime-dependent transmission mechanisms.
- Investigating adaptive feature representations and statistical learning methodologies to characterize cross-asset response heterogeneity and improve inference under non-stationary market conditions.

Stratified Non-Attendance Modeling in Clinical Datasets | *Dr. Fan Zengyan*

- Developing a unified statistical framework integrating survival analysis, generalized linear models, and robust inferential procedures to characterize heterogeneous non-attendance mechanisms.
- Incorporating heteroscedasticity-consistent covariance estimation, cluster-robust inference, and non-parametric variance decomposition to accommodate correlated observations and heterogeneous variability.
- Investigating demographic, behavioral, and temporal determinants of non-attendance through stratified estimation and model diagnostics, emphasizing inferential robustness across heterogeneous subpopulations.

Causal Machine Learning for Heterogeneous Treatment Effects | *Dr. Fan Zengyan*

- Developed a unified causal learning framework integrating Double Machine Learning and Causal Forests to estimate conditional average treatment effects in high-dimensional observational settings.
- Implemented cross-validated propensity score estimation, overlap diagnostics, and deterministic support trimming to improve estimation stability under limited covariate overlap.
- Applied unsupervised clustering to individualized treatment effect estimates to identify latent responder profiles and characterize heterogeneity in treatment responses.
- Constructed interpretable treatment assignment strategies using Optimal Policy Trees, linking heterogeneous effect estimation with individualized policy optimization.

Multivariate Time Series Forecasting for Equity Price Prediction | *Dr. Zhou Tianyi*

- Conducted a systematic empirical benchmark of classical statistical models and neural forecasting architectures under non-stationary, high-noise financial conditions.
- Evaluated ARIMA, ARIMAX, LSTM, N-HiTS, PatchTST, and StemGNN using walk-forward validation across multiple forecasting horizons to assess out-of-sample predictive performance.
- Investigated horizon-dependent model performance and the contribution of exogenous information through feature attribution analysis and comparative model evaluation.
- Established that parsimonious statistical models outperform more complex architectures at shorter horizons, while neural models exhibit stronger predictive performance over longer horizons.

Data-Driven Framework for Enhancing Supply Chain Resilience | *Dr. Liu Ning*

- Developed an end-to-end non-parametric learning framework for stochastic inventory control, integrating predictive representation learning with direct replenishment policy optimization.
- Formulated a decision-focused learning approach based on empirical risk minimization, circumventing explicit demand distribution modeling and conventional Predict-Then-Optimize pipelines.
- Incorporated feature attribution methodologies to investigate learned decision policies and quantify the influence of operational variables on replenishment decisions.
- Demonstrated improved inventory performance and robustness to distributional shift through systematic evaluation against conventional forecasting-based optimization baselines.

Compentencies

Programming & Scripting: Python, R, SQL, LaTeX
Software & Tools: Git, Jupyter, VS Code, Visual Studio
Languages: English, Mandarin